INCH-POUND

MIL-DTL-25707/1C
w/AMENDMENT 1
21 March 2022
SUPERSEDING
MIL-DTL-25707/1C
15 September 2011

DETAIL SPECIFICATION SHEET

# ANTENNA, FLUSH MOUNT, TYPE AT-740/A

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the antenna described herein shall consist of this document and MIL-DTL-25707.

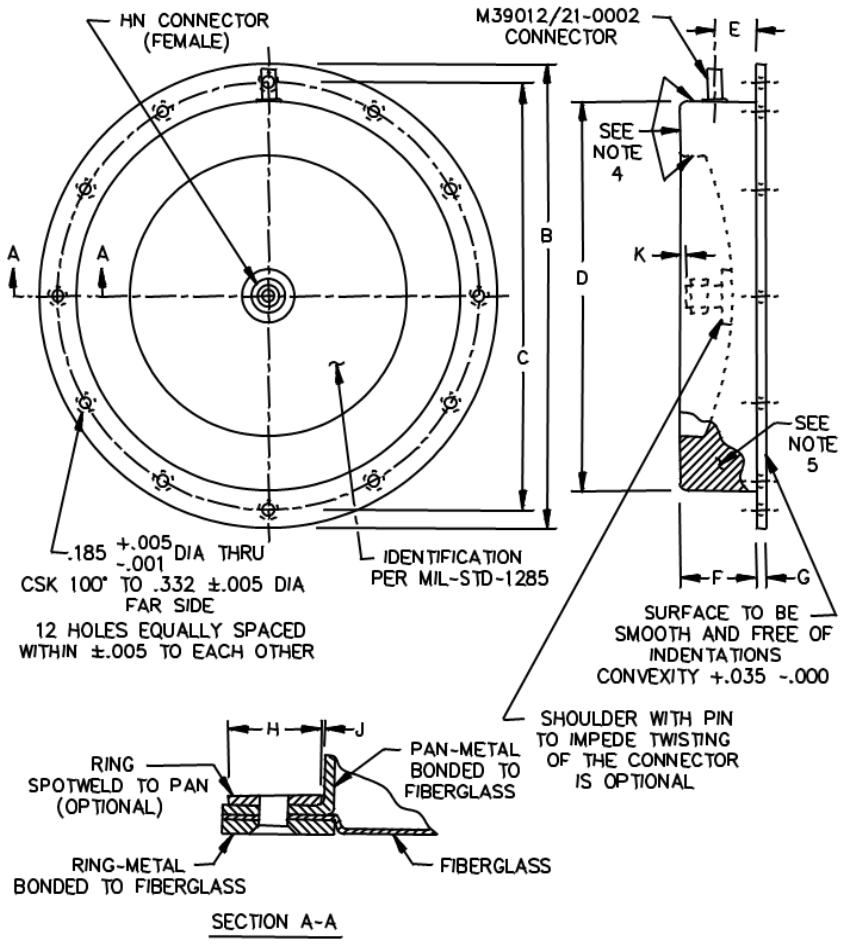



FIGURE 1. Outline drawing antenna AT-740/A.

AMSC N/A
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FSC 5985

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| Dimensions | | | | |
|---|---|---|---|---|
| Letter | Inches | Inches | Millimeters | Millimeters |
| | Min | Max | Min | Max |
| B | 8.391 | 8.421 | 213.13 | 213.89 |
| C | 7.709 | 7.729 | 195.81 | 196.32 |
| D | --- | 7.109 | --- | 180.57 |
| E | .750 | --- | 19.05 | --- |
| F | --- | 1.520 | --- | 38.61 |
| G | .141 | .156 | 3.58 | 3.96 |
| H | .60 | .72 | 15.24 | 18.29 |
| J | .00 | .12 | 0.00 | 3.05 |
| K | .06 | .13 | 1.52 | 3.30 |

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .015 inch (± 0.38 mm) for three place decimals and ± .03 inch (± 0.76 mm) for two place decimals.
4. Finish: Anodize ring in accordance with MIL-PRF-8625. Chemical conversion coating pan flange; paint pan body with flat white polyurethane enamel or equivalent; radome is to be flat black.
5. Fill antenna cavity with polyurethane foam, or equivalent, with density as necessary to maintain structural rigidity, radiation pattern as specified, and compliance to environmental requirements.

REQUIREMENTS:

Dimensions and construction: See figure 1.

Temperature operating range: -49°C to +96°C.

Frequency: 960 MHz to 1,220 MHz.

Impedance: 50 ohms, nominal.

VSWR of antenna: 960 MHz to 1,220 MHz, 1.8:1, maximum.
1,000 MHz to 1,100 MHz, 1.6:1, maximum.

VSWR of sampling probe: 960 MHz to 1,220 MHz, 1.5:1, maximum.

Probe attenuation: 17.5 dB ± 1.5 dB over frequency band 960 MHz to 1,220 MHz.

Power: 4 kW, peak; 100 watts, average.

Weight: Less than or equal to 16 ounces.

Part or Identifying Number (PIN): M25707/1-01.

Data list: The measured VSWR and attenuation data shall accompany each antenna when required by contract.



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Referenced documents. In addition to MIL-DTL-25707, this specification sheet references the following documents:
MIL-PRF-8625
MIL-STD-1285

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian:
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(Project 5985-2022-009)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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